

Anchorage and Fire-Blocking

Tying the building to its foundation, and stopping a fire traveling inside the walls. Four numbers, all asked directly.

Anchor bolts are what stop the whole timber structure sliding or lifting off the concrete. The exam asks for all three of their numbers, and it has asked you for two of them more than once.

The numbers here

Sole plate anchor bolt — diameter	1/2 in
Anchor bolt — minimum embedment into concrete or grouted masonry	7 in
Anchor bolt — maximum distance from each end of the plate section	12 in
Anchor bolts per plate section — minimum	2

YOU HAVE MISSED THIS 2 TIMES

Wood sole plates need to be anchored to foundation walls with _____ anchor bolts.
Half inch.

YOU HAVE MISSED THIS 2 TIMES

Foundation anchor bolts shall extend a minimum of _____ inches into concrete or grouted cells of masonry units.
Seven.

YOU HAVE MISSED THIS

There shall be a minimum of two anchor bolts per plate section with one bolt located not more than _____ inches from each end.
Twelve.

Read the set as one sentence: a half-inch bolt, seven inches deep, no more than twelve inches from each end of the plate, and never fewer than two per piece. Half, seven, twelve, two.

Fire-blocking

The numbers here

Fire-blocking — maximum horizontal concealed run in combustible construction	10 ft
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YOU HAVE MISSED THIS 2 TIMES

In combustible construction, what is the maximum horizontal measurement allowed in a concealed wall space between fire-blocking?

10 feet.

A concealed cavity is a chimney waiting to happen: fire gets into it and runs the length of the wall out of sight. Fire-blocking chops the cavity into lengths so the fire has to burn through something solid to continue. Ten feet horizontally is the limit.

Say it back

Cover the page. Answer each one out loud.

1. Say the four anchor bolt numbers as one sentence.
2. Anchor bolt diameter and embedment depth?
3. How close to the end of a plate section must a bolt sit, and how many per section?
4. Maximum horizontal concealed run between fire-blocking?
5. Explain in one sentence why fire-blocking exists.